CooLink - Marketplace Platform

Full-Stack Daily-Wage Worker Marketplace for India

Comprehensive Architecture, Implementation & Verification Report | October 2026

1. Executive Summary & Problem Solved

CooLink is a production-grade digital marketplace engineered to modernize the traditional daily-wage labor ecosystem in India. In the conventional system, unorganized workers (masons, electricians, carpenters, plumbers, painters) must travel early in the morning to street corners, junctions, or bus stands ("labor nakas") with no guarantee of work, leaving them vulnerable to wage exploitation, unpaid travel costs, and unsafe working arrangements.

Simultaneously, homeowners and civil contractors struggle with labor shortages, unpredictable attendance, lack of verified credentials, and no verified record of past performance. CooLink solves these foundational pain points by establishing a trust-first digital marketplace providing verified worker identities, real-time availability toggles, atomic booking concurrency, transparent wage baselines, and authenticated bi-directional reviews.

2. System Architecture & Technology Stack

The application is constructed with a clean decoupled architecture separating a RESTful API backend from a reactive, mobile-first single-page client:

- Backend Architecture: Node.js & Express with TypeScript, following strict Controller-Service-Repository separation with Zod request validation and centralized async error handling.
- Database & Persistence: MongoDB with Mongoose schemas. Features a dual-mode persistence layer: a zero-setup embedded MongoMemoryServer for instant zero-config evaluation, with seamless environment variable switching to MongoDB Atlas for production.
- Frontend Client: React 18 with TypeScript and Vite. Styled with Tailwind CSS for an ultra-fast, mobile-first responsive layout specifically tailored for one-handed smartphone use on construction job sites.
- Bilingual Accessibility: Instant 1-tap language switching between English and Malayalam (Malayalam script), reflecting grassroots regional usability in Kerala.
- Automated Test Suite: Comprehensive Jest & Supertest integration test suite covering identity authentication, atomic concurrency locks, and job review permissions (10/10 passing).

3. Security, Authentication & Verification Engine

Authentication and identity trust were prioritized over typical generic marketplace templates:

- Real OTP Phone Verification: Abstracted IOtpProvider pattern supporting production providers (Fast2SMS for Indian phone numbers, Twilio) alongside a clean DevLogger provider for zero-cost development.
- Dual-Token JWT Security: Short-lived access tokens (15m) paired with rotating refresh tokens (7d). Protected routes feature an Axios response interceptor that transparently refreshes expired sessions without interrupting the user.
- Role-Based Access Control (RBAC): Strict separation between Worker, Employer (Contractor / Home Owner), and Platform Admin roles, enforced via server-side middleware.
- Government ID Verification (KYC): Aadhaar / Voter ID / Driving License submission flow. Documents and ID numbers are encrypted and isolated; never exposed publicly. Profiles display verified status badges only after administrator approval.

- Location Privacy Protection: Exact worker home addresses are never exposed publicly. Employer discovery queries compute distance while displaying only fuzzy locality circles (~2km radius) to protect worker privacy.

4. Core Features & Business Workflows

A. Worker One-Tap Availability Toggle

Workers can immediately broadcast their work status with a single tap on their dashboard: "Available Today", "Available Tomorrow", "Available from Date", or "Busy / Not Available". When a worker accepts a confirmed job, the system atomically locks their status to BUSY, preventing overlapping commitments.

B. Deterministic Job Lifecycle State Machine

Jobs strictly transition through validated states: POSTED -> WORKER_REQUESTED -> ACCEPTED -> IN_PROGRESS -> COMPLETED -> RATED. Invalid state jumps (such as marking a completed job back to posted) are rejected by schema validators.

C. Concurrency & Anti-Double-Booking Engine

To eliminate race conditions when multiple contractors attempt to hire the same worker simultaneously, booking requests use atomic MongoDB conditional updates with session isolation, ensuring zero double bookings.

D. Authenticated Bi-Directional Reviews

Unlike generic public review boards, ratings (1 to 5 stars) and feedback comments can ONLY be submitted by participants of a verifiably COMPLETED job. Workers can review contractors on payment promptness and site safety, while employers review worker craftsmanship.

5. Seeded Kerala Data & Test Accounts

The system includes realistic seed profiles localized for Kerala (Malappuram, Kottakkal, Tirur, Calicut) with verified skills, wage histories, and pending work requests:

Role	Name	Phone Number	Password	Key Details
Worker	Ravi Kumar	9847100001	password123	Verified Mason (Rs. 950/day)
Worker	Suresh Babu	9847100002	password123	Verified Tile Worker (Rs. 1,000/day)
Employer	Moideen Kutty	9847000001	password123	Contractor (Kottakkal Builders)
Employer	Dr. Anita Thomas	9847000002	password123	Home Owner (Malappuram)
Admin	Platform Admin	9847000000	adminpassword123	KYC & Moderation Panel

6. How to Run the Project

Quickstart Commands

Backend Server: cd server && npm run dev (Runs on http://localhost:5000)
Frontend Client: cd client && npm run dev (Runs on http://localhost:5173)
Automated Tests: cd server && npm test (Executes 10 unit and integration tests)
Database Note: Starts automatically in-memory. Zero manual database installation required.

7. Future Roadmap & Production Scaling

- Live SMS Integration: Switch SMS_PROVIDER=FAST2SMS in server/.env with an active DLT-approved sender ID for live OTP delivery.
- Escrow Payments & UPI: Integration of Razorpay Route or Cashfree UPI escrow to lock wages upon job start and release upon mutual completion confirmation.

- Deployment Pipeline: Single-click containerized deploy with Docker, backend hosted on Render/Railway with MongoDB Atlas, and frontend on Vercel.